Midterm Jeopardy Review

Module 1, 2, 3

ABOUT ANDRE

$1

$2

What does Andre like to do for fun:

a) Build legos

b) LARP

c) Hike

d) Read programming books

SHOW ANSWER

What does Andre like to do for fun:

a) Build legos

b) LARP

c) Hike

d) Read programming books

BACK TO BOARD

What is Andre's favorite programming language?

a) Java

b) Go

c) Scheme

d) Javascript

SHOW ANSWER

What is Andre's favorite programming language?

a) Java

b) Go

c) Scheme

d) Javascript

BACK TO BOARD

MACHINES & HISTORY	PHASES OF COMPILATION	LANGUAGE THEORY, REGEX & SCANNERS	GRAMMARS & PARSERS	THE STACK & THE HEAP	SCOPE, BINDING & NAMES
$100	$100	$100	$100	$100	$100
$200	$200	$200	$200	$200	$200
$300	$300	$300	$300	$300	$300
$400	$400	$400	$400	$400	$400
$500	$500	$500	$500	$500	$500

MACHINES & HISTORY | $100

True or False: On the mechanical calculator, entering '11' and pressing '+' eleven times results in 121.

SHOW ANSWER

True or False: On the mechanical calculator, entering '11' and pressing '+' eleven times results in 121.

BACK TO BOARD

MACHINES & HISTORY | $200

The calculator stored its data in _____ and could perform simple operations like add, subtract, multiply, and divide.

SHOW ANSWER

The calculator stored its data in __registers__ and could perform simple operations like add, subtract, multiply, and divide.

BACK TO BOARD

Who is widely considered the author of the world's first computer program, written for Babbage's analytical engine?

a) Grace Hopper

b) Ada Lovelace

c) Alan Turing

d) John von Neumann

SHOW ANSWER

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BACK TO BOARD

MACHINES & HISTORY | $400

List the 5 components of the von Neumann architecture.

SHOW ANSWER

List the 5 components of the von Neumann architecture.

Input Device, Output Device, Control Unit, ALU, Memory

BACK TO BOARD

What would have to be done to add a NEW function, such as square root, to the mechanical calculator?

SHOW ANSWER

What would have to be done to add a NEW function, such as square root, to the mechanical calculator?

Mechanical Redesign

BACK TO BOARD

PHASES OF COMPILATION | $100

The scanner divides the program into _____, the smallest meaningful units in a translator.

SHOW ANSWER

The scanner divides the program into **tokens**, the smallest meaningful units in a translator.

BACK TO BOARD

Parsing is recognition of what kind of language?

a) Regular

b) Context-free

c) Context-sensitive

d) Unambiguous

SHOW ANSWER

Parsing is recognition of what kind of language?

a) Regular

b) Context-free

c) Context-sensitive

d) Unambiguous

BACK TO BOARD

PHASES OF COMPILATION | $300

True or False: Detecting an array subscript out of bounds is always possible at compile time.

SHOW ANSWER

True or False: Detecting an array subscript out of bounds is always possible at compile time.

BACK TO BOARD

PHASES OF COMPILATION | $400

What is the input and output of the Scanner?
What is the input and output of the Parser?

SHOW ANSWER

What is the input and output of the Scanner?

Character Stream, Token Stream

What is the input and output of the Parser?

Token Stream, Parse Tree

BACK TO BOARD

PHASES OF COMPILATION | $500

Nina

SHOW ANSWER

PHASES OF COMPILATION | $500 | ANSWER

BACK TO BOARD

True or False: Syntax is the set of rules that specify what a program does when it executes.

SHOW ANSWER

True or False: Syntax is the set of rules that specify what a program does when it executes.

BACK TO BOARD

Which pairing of compiler part and rule notation is correct?

a) Scanner, CFG, push-down automaton

b) Scanner, regular expressions, DFA; Parser, CFG, push-down automaton

c) Scanner -- CFG -- DFA

d) Parser -- regular expressions -- DFA

SHOW ANSWER

Which pairing of compiler part and rule notation is correct?

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c) Scanner -- CFG -- DFA

d) Parser -- regular expressions -- DFA

BACK TO BOARD

Using the following RE:

```
integer: digit digit*
decimal: digit* (. digit | digit .) digit*
exponent: (e | E) (+ | - | epsilon) integer)
```

which string is NOT a valid number?

a) 007

b) .5

c) 1E+10

d) -7

SHOW ANSWER

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BACK TO BOARD

What does this shell command output?

```
$ echo "banana" | sed 's/a/o/g'
```

SHOW ANSWER

What does this shell command output?

```
$ echo "banana" | sed 's/a/o/g'
$ bonono
```

BACK TO BOARD

What does a scanner do with source code comments?

SHOW ANSWER

What does a scanner do with source code comments?

Ignores them.

BACK TO BOARD

LL stands for "Left to right, _____ derivation."

SHOW ANSWER

LL stands for "Left to right, left-most derivation."

BACK TO BOARD

Which parsers are also known as bottom-up or shift-reduce parsers?

a) LL parsers

b) LR parsers

c) Earley parsers

d) Predictive parsers

SHOW ANSWER

Which parsers are also known as bottom-up or shift-reduce parsers?

a) LL parsers

b) LR parsers

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BACK TO BOARD

True or False: A regular expression cam match properly nested parentheses.

SHOW ANSWER

True or False: A regular expression can match properly nested parentheses.

BACK TO BOARD

Nina needs to show that a given CFG is ambiguous. How can she prove it?

SHOW ANSWER

Nina needs to show that a given CFG is ambiguous. How can she prove it?

Choose a string and show two unique parse trees.

BACK TO BOARD

The string "10 + 4 + 3" has two different parse trees using the CFG:

```
expr -> id | number | - expr | ( expr ) | expr op expr
```

Show both.

SHOW ANSWER

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```
expr -> id | number | - expr | ( expr ) | expr op expr
```

Show both.

BACK TO BOARD

Each time a subroutine is called, its data is pushed on the stack grouped into a _____.

SHOW ANSWER

Each time a subroutine is called, its data is pushed on the stack grouped into a _frame_ .

BACK TO BOARD

Which of these is NOT typically stored in a function's stack frame?

a) Local variables

b) Function name

c) Formal parameters

d) Return address

SHOW ANSWER

Which of these is NOT typically stored in a function's stack frame?

a) Local variables

b) Function name

c) Formal parameters

d) Return address

BACK TO BOARD

Internal fragmentation occurs when:

a) Freed blocks leave gaps between active objects

b) Allocated memory is larger than required for an object

c) The stack and heap collide

d) A pointer is never freed

SHOW ANSWER

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a) Freed blocks leave gaps between active objects

b) Allocated memory is larger than required for an object

c) The stack and heap collide

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BACK TO BOARD

THE STACK & THE HEAP | $400

Give one argument for garbage collection and one argument against it.

SHOW ANSWER

Give one argument for garbage collection and one argument against it.

BACK TO BOARD

New objects can be allocated in C with malloc or in Java with new. How many new objects does each call create, and where are they allocated?

```
int *a = (int *)malloc(ARRAY_SIZE * sizeof(int));
```

```
Object obj = new Object();
```

SHOW ANSWER

New objects can be allocated in C with malloc or in Java with new. How many new objects does each call create, and where are they allocated?

Two objects each. One on stack and one on heap.

BACK TO BOARD

Define shadowing.
Define aliasing.

SHOW ANSWER

Define shadowing: same name, different objects

Define aliasing: same object, different names

BACK TO BOARD

Static scope is also called:

a) Runtime scope

b) Lexical scope

c) Heap scope

d) Global scope

SHOW ANSWER

Static scope is also called:

a) Runtime scope

b) Lexical scope

c) Heap scope

d) Global scope

BACK TO BOARD

True or False: In Java, only classes can be parameterized with different types.

SHOW ANSWER

True or False: In Java, only classes can be parameterized with different types.

BACK TO BOARD

A subroutine uses the bindings in effect when it is created. This is called:

a) Deep binding

b) Shallow binding

c) Static binding

d) Overloading

SHOW ANSWER

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a) Deep binding

b) Shallow binding

c) Static binding

d) Overloading

BACK TO BOARD

Explain why higher-order.js does not print 3 after variable is incremented.

SHOW ANSWER

Explain why higher-order.js does not print 3 after variable is incremented.
Closure property.

BACK TO BOARD